

WORKED EXAMPLE

Two different questions about SGD

How fast can an objective gap shrink, and how noisy is a sample mean?

Part A works through the step-size conditions and rate expressions in Lecture5 pages56-58 and86.

Part B calculates the variance of sampled losses from pages65-66 and85.

All constants and tiny distributions below are chosen study examples. The rate calculations use stipulated mathematical bounds, not measured predictions for your neural network.

An update changes parameters once. An example-gradient evaluation computes one training example contribution. These counts differ for full batches, SGD and mini-batches.

For the sampling calculations, freeze the parameters first. Otherwise the loss distribution itself changes while you measure it.

WORKED EXAMPLE

Read the two infinite-sum conditions

Lecture5 page56; index k starts at 1 in this lesson.

$$\eta_k = \frac{c}{k^p}, \quad c > 0, \quad k = 1, 2, 3, \dots$$

$$\sum_{k=1}^{\infty} \eta_k = \infty, \quad \sum_{k=1}^{\infty} \eta_k^2 < \infty$$

An infinite sum is the limiting total of more and more terms. The first condition says accumulated step sizes never reach a finite ceiling; the second says accumulated squared step sizes do.

These are step-size conditions within convergence theorems. Such theorems also require suitable objective regularity, controlled iterates and gradient-noise assumptions, such as unbiased estimates with bounded variance.

A convex objective supports conclusions different from a general nonconvex neural-network loss. These two sums alone do not guarantee a local minimum or global search.

WORKED EXAMPLE

Apply the p-series test to both sums

The series $1/k$ raised to exponent a has a finite sum exactly when $a > 1$.

$$\sum \eta_k = c \sum \frac{1}{k^p} \implies \text{diverges if } p \leq 1$$

$$\sum \eta_k^2 = c^2 \sum \frac{1}{k^{2p}} \implies \text{finite if } 2p > 1$$

$$2p > 1 \implies p > \frac{1}{2}; \quad \frac{1}{2} < p \leq 1$$

Multiplying a positive series by the finite positive constant c or c squared does not change whether it converges. Both tests must pass at the same time.

The lower endpoint is excluded; the upper endpoint is included.

WORKED EXAMPLE

Check the endpoints instead of guessing

A smaller rate is not automatically a suitable rate.

p	sum of rates	sum of squared rates	both?
0.5	exponent .5: infinite	exponent 1: infinite	no
0.75	exponent .75: infinite	exponent 1.5: finite	yes
1	exponent 1: infinite	exponent 2: finite	yes
2	exponent 2: finite	exponent 4: finite	no

$$p = 1: \quad \eta_k = c/k, \quad \eta_k^2 = c^2/k^2$$

At $p=.5$ the squared sequence is harmonic, so its sum still diverges. At $p=2$ the rates shrink so quickly that their total is finite.

WORKED EXAMPLE

Geometric shrinking fails the first sum

Choose $\eta_k = c(0.9)^k$ with k starting at 1.

$$\sum_{k=1}^{\infty} c(0.9)^k = \frac{0.9c}{1-0.9} = 9c < \infty$$

$$\sum_{k=1}^{\infty} c^2(0.81)^k = \frac{0.81c^2}{1-0.81} = \frac{81}{19}c^2$$

Both totals are finite. The square-sum condition passes, but the required divergent sum of rates fails.

This does not say geometric decay is never useful in finite training. It says this schedule does not meet this particular pair of infinite-horizon conditions.

WORKED EXAMPLE

Turn an inverse gap bound into an update count

Define gap as objective value minus its optimal value.

$$\text{gap}_k = J(w_k) - J^*, \quad \text{gap}_k \leq \frac{2}{k}$$

$$\frac{2}{k} \leq 0.01 \implies 2 \leq 0.01k \implies k \geq 200$$

$$k = 199 : 2/199 \approx 0.0100503 > 0.01$$

$$k = 200 : 2/200 = 0.01$$

J^* is the best objective value in the stated optimization problem. The bound guarantees the requested tolerance from update 200 onward under its assumptions; actual convergence could be earlier.

WORKED EXAMPLE

Solve a geometric bound using logarithms

Assume instead $\text{gap}_k \leq 2(0.8)^k$.

$$2(0.8)^k \leq 0.01 \implies (0.8)^k \leq 0.005$$

$$k \ln(0.8) \leq \ln(0.005)$$

$$k \geq \frac{\ln(0.005)}{\ln(0.8)} \approx 23.7432 \implies k_{\min} = 24$$

Taking natural logs preserves the inequality because log increases. Dividing by $\ln(0.8)$, which is negative, reverses the inequality. Round up to a whole update.

$$2(0.8)^{23} \approx 0.0118059, \quad 2(0.8)^{24} \approx 0.00944473$$

WORKED EXAMPLE

Fewer updates can still mean more example work

Illustration: assign the inverse bound to SGD and geometric bound to full gradient.

$T = 100$ training examples

SGD : $200 \text{ updates} \times 1 = 200 \text{ example gradients}$

full : $24 \text{ updates} \times 100 = 2400 \text{ example gradients}$

T is dataset size. These counts follow the stipulated bounds and cost model; they are not a measured comparison or a claim that one optimizer always wins.

Vectorized and parallel computations change elapsed time. Equal update counts, equal example work and equal runtime are three different comparisons.

WORKED EXAMPLE

Rewrite the mini-batch expression at equal work

Lecture5 page86 gives $O(1/\sqrt{bk}) + 1/k$.

$$R(b, k) = \frac{1}{\sqrt{bk}} + \frac{1}{k}$$

Big-O describes scale up to constants in a stated regime. For arithmetic only, R sets both hidden coefficients to 1; it is not an exact measured gap or a universal numerical bound.

$$M = bk \implies k = M/b$$

$$R = \frac{1}{\sqrt{b(M/b)}} + \frac{1}{M/b} = \frac{1}{\sqrt{M}} + \frac{b}{M}$$

b is batch size, k is updates, and M is example-gradient work. The substitution keeps total work fixed when batch size changes.

WORKED EXAMPLE

Evaluate both terms separately

Hold $M=10,000$ example-gradient evaluations fixed.

$$b = 25 : \quad k = 10000/25 = 400$$

$$R = 1/\sqrt{10000} + 25/10000 = 0.01 + 0.0025 = 0.0125$$

$$b = 100 : \quad k = 10000/100 = 100$$

$$R = 1/\sqrt{10000} + 100/10000 = 0.01 + 0.01 = 0.02$$

The first term is unchanged at equal M ; the second grows with b . This expression alone does not establish the slide's blanket square-root-of- b degradation in total work, nor a hardware-time prediction.

WORKED EXAMPLE

A tiny population of losses

Keep parameters fixed and draw a fresh example independently.

$$P(D = 1) = P(D = 3) = \frac{1}{2}$$

D is the random loss from one example. Expectation is its probability-weighted mean. Variance is its mean squared distance from that mean.

$$\mu = \mathbb{E}[D] = \frac{1}{2}(1) + \frac{1}{2}(3) = 2$$

$$\sigma^2 = \text{Var}(D) = \frac{1}{2}(1 - 2)^2 + \frac{1}{2}(3 - 2)^2 = 1$$

$$\sigma = \sqrt{\sigma^2} = 1$$

Standard deviation is the square root of variance and has the same units as the loss. Variance has squared units.

WORKED EXAMPLE

Enumerate every two-draw mini-batch

Independent draws allow repeats; each ordered pair has probability 1/4.

D1	D2	mean (D1+D2)/2	squared distance from 2
1	1	1	1
1	3	2	0
3	1	2	0
3	3	3	1

$$E[\bar{D}] = (1 + 2 + 2 + 3)/4 = 2$$

$$\text{Var}(\bar{D}) = (1 + 0 + 0 + 1)/4 = 0.5$$

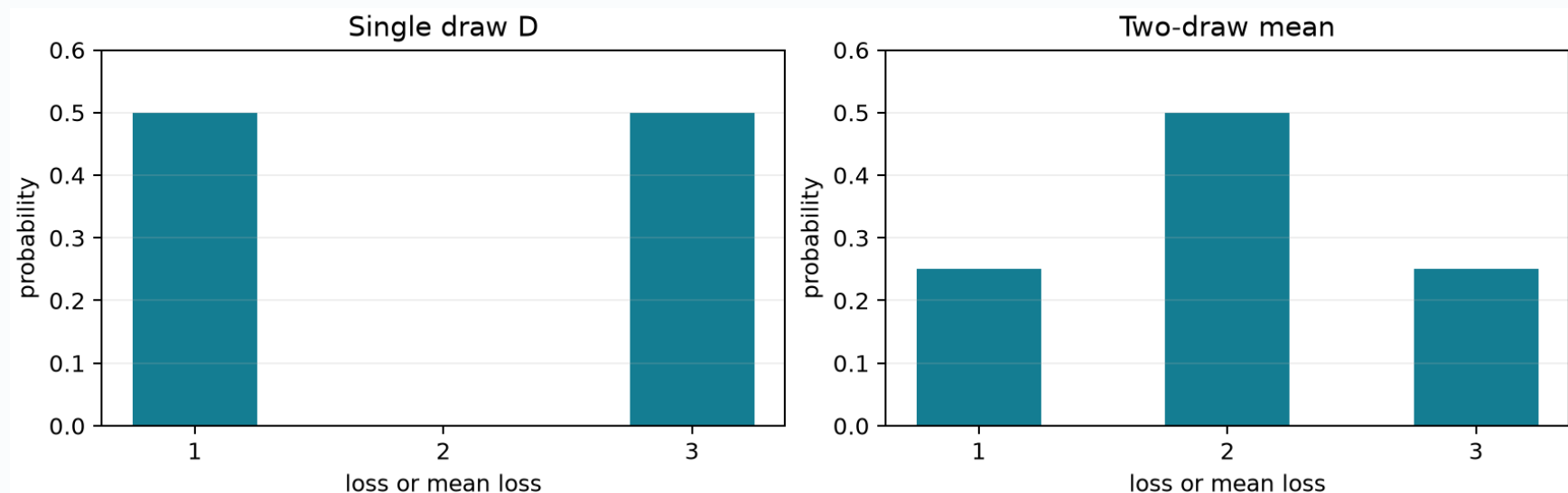
$$\text{SD}(\bar{D}) = \sqrt{0.5} \approx 0.707107$$

The mean remains 2 but the sample mean varies less. Variance .5 is not standard deviation .5.

WORKED EXAMPLE

See how averaging concentrates probability

This is a sampling distribution at fixed parameters, not a training curve.



A single draw never equals 2. The mean of two draws equals 2 half the time. It can still be 1 or 3, but those outcomes become less likely.

WORKED EXAMPLE

Why independent batch variance divides by b

Every draw has the same variance sigma squared and is independent of the others.

$$\bar{D} = \frac{1}{b} \sum_{i=1}^b D_i$$

$$\text{Var}(\bar{D}) = \frac{1}{b^2} \text{Var}\left(\sum_{i=1}^b D_i\right)$$

$$= \frac{1}{b^2} \sum_{i=1}^b \sigma^2 = \frac{b\sigma^2}{b^2} = \frac{\sigma^2}{b}$$

Scaling by 1/b scales squared deviations by 1/b squared. Covariance measures how deviations of two draws vary together. Independence makes these cross terms zero, so variances add.

$$\mathbb{E}[\bar{D}] = \mu, \quad \text{SD}(\bar{D}) = \frac{\sigma}{\sqrt{b}}$$

WORKED EXAMPLE

Calculate variance and standard deviation separately

Suppose a single-example loss has variance 9.

$$\sigma^2 = 9 \implies \sigma = \sqrt{9} = 3$$

$$b = 9 : \quad \text{Var}(\bar{D}) = 9/9 = 1, \quad \text{SD} = \sqrt{1} = 1$$

$$b = 100 : \quad \text{Var}(\bar{D}) = 9/100 = 0.09$$

$$\text{SD} = \sqrt{0.09} = 0.3$$

A hundred independent draws reduce variance by 100 and standard deviation by 10. These are different factors because one measure is squared.

WORKED EXAMPLE

Without replacement is a different experiment

The complete fixed dataset is [1,3]; select both members.

$$\bar{D} = (1 + 3)/2 = 2 \quad \text{every time}$$

$$\text{Var}(\bar{D}) = 0$$

The earlier independent draws allowed (1,1) and (3,3). Selecting both distinct dataset members does not. There is no remaining randomness in the mean.

$$\sigma_{pop}^2 = ((1 - 2)^2 + (3 - 2)^2)/2 = 1$$

The population variance here uses divisor N=2. This is the spread of the fixed dataset, not an estimated sample variance with divisor N-1.

OPTIONAL EXTENSION

Optional: finite-population correction

For a uniform subset of b distinct examples from a fixed dataset of size $N > 1$.

$$\text{Var}(\bar{D}) = \frac{\sigma_{pop}^2}{b} \frac{N-b}{N-1}$$

$$N = 2, b = 2 : \quad \frac{1}{2} \frac{2-2}{2-1} = 0$$

$$N = 2, b = 1 : \quad \frac{1}{1} \frac{2-1}{2-1} = 1$$

This supporting sampling formula is an extension. The factor $(N-b)/(N-1)$ accounts for dependence caused by sampling without replacement.

Unbiased risk estimation at a fixed parameter does not imply unbiased test performance after choosing parameters using those same data. Training changes parameters and selection uses the observed losses.

INDEPENDENT PRACTICE

Your turn: schedules and bound arithmetic

Solve before reading the answer pages.

For $\eta_k = c/k^p$ with $k \geq 1$, test $p = .4, .6, 1, 1.1$ against both infinite-sum conditions. Show the exponent of each squared sequence.

$$\text{gap} \leq 3/k \quad \text{or} \quad \text{gap} \leq 3(0.5)^k$$

Under each separate stipulated bound, find the smallest whole update count sufficient for $\text{gap} \leq .03$. Check the preceding update.

Evaluate the illustrative $R = 1/\sqrt{M} + b/M$ with $M = 1600$ and $b = 16$. Show k and each term.

INDEPENDENT PRACTICE

Your turn: a fresh loss distribution

Keep parameters fixed; loss_0 and loss_4 each have probability $1/2$.

Calculate the mean, variance and standard deviation of one draw, and of the mean of four independent draws.

For a direct check, let j be the number of loss_4 draws out of four. The 16 equally likely ordered outcomes have counts 1,4,6,4,1 for $j=0,1,2,3,4$. Calculate each sample mean and then its variance.

If the entire fixed dataset $[0,4]$ is selected without replacement, what is the mean and its variance? Explain the difference from independent draws.

WORKED ANSWER

Answer: test each exponent twice

The accepted interval is $1/2 < p \leq 1$.

p	$p \leq 1$?	squared exponent $2p > 1$?	both
.4	yes	.8: no	no
.6	yes	1.2: yes	yes
1	yes	2: yes	yes
1.1	no	2.2: yes	no

$p = 0.6$ and $p = 1$

At $p=.4$ the first sum diverges as required, but the squared sum also diverges. At $p=1$ the squared sum converges, but the rate sum also converges.

WORKED ANSWER

Answer: inverse and geometric counts

These are sufficient counts under the two supplied bounds.

$$3/k \leq 0.03 \implies k \geq 3/0.03 = 100$$

$$3/99 \approx 0.030303 > 0.03, \quad 3/100 = 0.03$$

$$3(0.5)^k \leq 0.03 \implies (0.5)^k \leq 0.01$$

$$k \geq \ln(0.01)/\ln(0.5) \approx 6.64386 \implies k_{min} = 7$$

$$3(0.5)^6 = 0.046875, \quad 3(0.5)^7 = 0.0234375$$

Again, division by a negative logarithm reverses the inequality. The integer ceiling means round up.

WORKED ANSWER

Answer: equal work and one-draw variability

Keep arithmetic for work counts separate from sampling variability.

$$k = M/b = 1600/16 = 100$$

$$R = 1/\sqrt{1600} + 16/1600 = 1/40 + 0.01 = 0.035$$

$$\mu = 0.5(0) + 0.5(4) = 2$$

$$\sigma^2 = 0.5(0 - 2)^2 + 0.5(4 - 2)^2 = 4$$

$$\sigma = \sqrt{4} = 2$$

The loss distribution has the same mean² as the main example but twice its standard deviation.

WORKED ANSWER

Answer: four independent draws

If j draws have loss 4, the mean is $(4j)/4=j$.

j / sample mean	number of outcomes	probability	(mean-2)^2
0	1	1/16	4
1	4	4/16	1
2	6	6/16	0
3	4	4/16	1
4	1	1/16	4

$$E[\bar{D}] = (0 + 4 + 12 + 12 + 4)/16 = 2$$

$$\text{Var}(\bar{D}) = (4 + 4 + 0 + 4 + 4)/16 = 1$$

$$\text{SD} = 1 \quad \text{also : } \sigma^2/b = 4/4 = 1$$

WORKED ANSWER

Answer: a full dataset has no sampling uncertainty

Without replacement, taking both members of $[0,4]$ fixes its mean.

$$\bar{D} = (0 + 4)/2 = 2, \quad \text{Var}(\bar{D}) = 0$$

A fixed full-dataset mean can still differ from the population risk. Zero randomness from selecting this entire dataset is not a promise of zero estimation error relative to the real world.

Exam checklist: identify update versus example count; keep the logarithm inequality direction correct; distinguish a stated bound from big-O scale; name the sampling procedure; freeze parameters; take a square root only when asked for standard deviation.